

QoR Surrogate Modeling for Early HLS-to-ASIC Decision Making

Can final OpenROAD QoR be estimated from information available before backend physical implementation?

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Early signal
→
Backend QoR

Motivation: reduce backend effort, not replace physical design

Design-space exploration produces many HLS configurations, while final place and route is the expensive stage.

Question

Can early features predict final area and timing well enough to choose backend candidates?

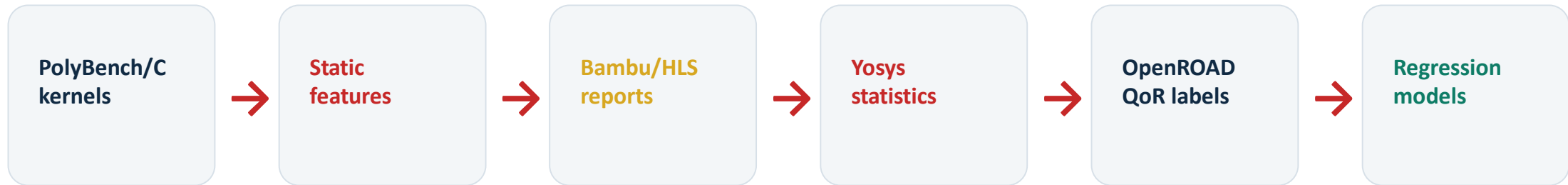
Intended use

- rank candidate configurations
- filter weak options early
- prioritize backend runs

Surrogate = Decision support. OpenROAD remains the final physical-design authority.

Implemented source-to-backend workflow

Parsing and modeling stages are implemented as Python scripts in the repository.



Static parsing: `extract_static_features.py` | Yosys: `parse_yosys_reports.py` | OpenROAD: `parse_openroad_reports.py`

Modeling: `train_baseline.py` · `train_feature_engineered_models.py` · `train_rigorous_feature_models.py` · `train_ablation_models.py`

Only features available before final OpenROAD are used for the main surrogate inputs.

Dataset progression: 864 to 79 to 18



The supervised regression problem is deliberately small because final backend outputs are the expensive labels.

Interpret results as an initial feasibility study - not a production predictor.

Early inputs, final targets, and leakage prevention

Valid pre-OpenROAD feature groups

- configuration: clock, dataset size, memory policy, DSP, optimization
- static C: lines, loops, branches, accesses, operations
- Bambu/HLS: control steps, states, area estimate, registers, mux area
- Yosys: wires, wire bits, cells, memories, processes, modules
- derived ratios from the same early information

Prediction targets

Final ASIC area
Critical-path delay
Total power (exploratory)

Leakage rule

Post-routing: analysis only.
Violations, DEF wirelength,
route guides, DRC.

Validation tests generalization to unseen benchmark kernels

Models evaluated

- Linear / Ridge / Elastic Net
- Random Forest / Gradient Boosting
- Gaussian Process / PLS / PCA variants
- Feature-group comparisons

Main evaluation: GroupKFold by benchmark

TRAIN

all non-held-out benchmark kernels

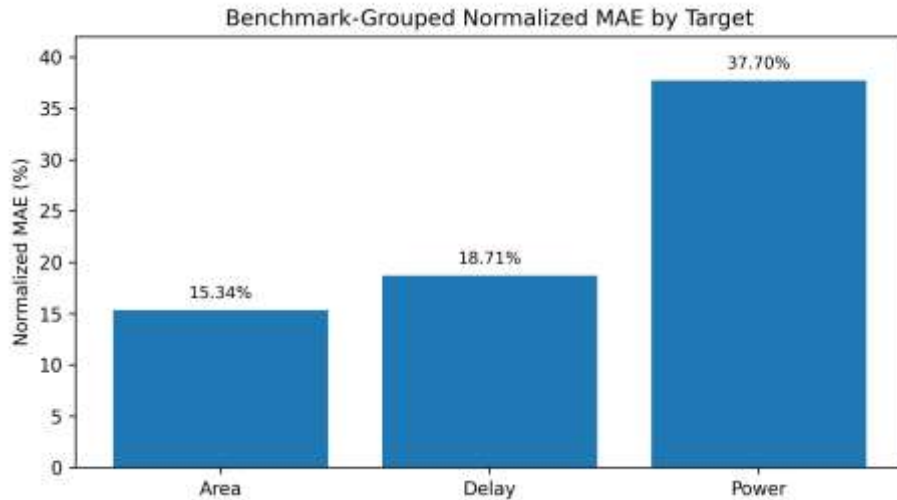
TEST

one complete
benchmark
kernel

Random splits can share benchmark-specific structure across train/test and therefore look optimistic.

NMAE = MAE / mean(|y|) → normalized prediction error, not accuracy

GroupKFold results: area strongest, power exploratory



Report Figure 1 - normalized MAE

15.34%

Area NMAE
MAE: 6637.11

18.71%

Delay NMAE
MAE: 0.763 ns

37.70%

Power NMAE
MAE: 0.0157 W

Best setup

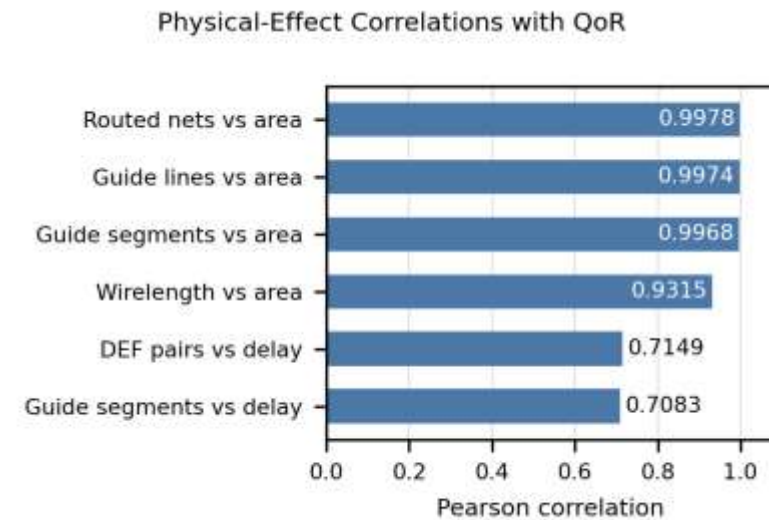
Elastic Net
raw pre-OpenROAD

Elastic Net
derived ratios

Elastic Net
Yosys-only

Power: exploratory - skewed and clock-sensitive.

Physical effects: post-routing analysis only



Report Figure 2 - post-routing correlations

Observed artifacts and closure

18 DEF + route guide + DRC reports
Setup / hold / max-slew: 0 violations
Max capacitance: 3 total, 2 max/design

Selected correlations

Area: routed nets 0.9978 · guide nonempty 0.9974
Area: guide segments 0.9968 · approx. wirelength 0.9315
Delay: DEF pairs 0.7149 · guide segments 0.7083

Wirelength is Manhattan-distance approximation from DEF coordinate sequences - not official router wirelength. DRC: 12 clean/empty, 6 possible violations, 7 keyword hits (max 2).

Limitations define the next experiments

Current limitations

- only 18 complete OpenROAD labels
- one aggressive-clock run and limited power diversity
- no systematically measured wall-clock speed-up
- static extraction is lightweight pattern-based parsing

Most useful next steps

- add backend labels systematically across configurations and constraints
- select runs with active learning or uncertainty sampling
- improve static features with AST / LLVM analysis
- enrich Yosys and pre-routing complexity proxies

Power: use transformations, robust models, clock-normalized targets, or separate clock regimes.

Conclusion: useful early signal, measured scope

A complete, reproducible source-to-backend modeling pipeline was implemented.

1 Useful early signal

Area: 15.34% NMAE
Delay: 18.71% NMAE

2 Realistic evaluation

GroupKFold holds out
complete benchmarks

3 Appropriate scope

Feasibility study;
not a replacement for
OpenROAD

Main next step: more diverse OpenROAD labels for robustness.